

Answer Key: sist

Course: SIS102

Part I: MCQ Answers

1. Answer: A
2. Answer: B
3. Answer: A
4. Answer: C
5. Answer: C
6. Answer: B
7. Answer: C
8. Answer: D
9. Answer: D
10. Answer: D
11. Answer: A
12. Answer: D
13. Answer: C
14. Answer: B
15. Answer: D
16. Answer: A
17. Answer: C
18. Answer: C
19. Answer: B

Part II: True / False Answers

1. Answer: False

Statement: The most elusive threats to an organization come from outside the organization.

2. Answer: True

Statement: Insider threats can be caused by malicious intent.

3. Answer: False

Statement: Ransomware is a type of malicious software that steals data.

4. Answer: True

Statement: Ransomware attackers often threaten to release sensitive stolen data if the ransom is not paid.

5. Answer: True

Statement: Regular software updates can help protect against ransomware.

6. Answer: True

Statement: Educating employees about phishing can help prevent insider threats.

7. Answer: True

Statement: Immutable storage architectures can help protect against ransomware.

8. Answer: True

Statement: Application whitelisting can help prevent ransomware attacks.

9. Answer: True

Statement: Network segmentation can help prevent the spread of ransomware.

10. Answer: True

Statement: The Defense in Depth strategy combines general malware defenses with ransomware-specific tactics.

Part III: Written Answers

1. What are the causes of insider threats?

Answer: The causes of insider threats include malicious intent, mistakes, and negligence. These causes can lead to intentional or unintentional harm to the organization.

2. How can insider threats be prevented?

Answer: Insider threats can be prevented through a combination of technical controls and organizational measures. Technical controls include access management and user behavior analytics, while organizational measures include regular training and a culture of security awareness.

3. What is ransomware and how does it work?

Answer: Ransomware is a type of malicious software that encrypts files on a device or network, holding them hostage until a ransom is paid. It typically infiltrates systems through phishing emails, exploiting vulnerabilities, or malicious downloads.

4. What are some ways to protect against ransomware?

Answer: Some ways to protect against ransomware include regular software updates, robust endpoint security solutions, and educating employees about phishing. Additionally, specific ransomware countermeasures such as immutable storage architectures, application whitelisting, and network segmentation can also be effective.

5. What is the Defense in Depth strategy and how does it relate to ransomware protection?

Answer: The Defense in Depth strategy combines general malware defenses with ransomware-specific tactics to provide comprehensive protection against ransomware attacks. This approach includes using multiple layers of defense, such as firewalls, intrusion detection systems, and encryption, to prevent, detect, and respond to ransomware attacks.

6. What is the purpose of threat intelligence services in relation to ransomware protection?

Answer: Threat intelligence services provide information about potential threats, including ransomware, to help organizations prepare and respond to attacks. This information can be used to inform security decisions and improve incident response plans.

7. How can end-to-end encryption help protect against ransomware?

Answer: End-to-end encryption can help protect against ransomware by encrypting data in transit, making it more difficult for attackers to intercept and exploit sensitive information. This can help prevent data breaches and reduce the risk of ransomware attacks.

8. What is incident response and how does it relate to ransomware protection?

Answer: Incident response is a plan to respond to and manage the aftermath of a security incident, including a ransomware attack. It involves identifying the incident, containing the damage, and restoring systems and data to a secure state.

9. What is threat hunting and how does it relate to ransomware protection?

Answer: Threat hunting is the process of identifying and mitigating potential threats, including ransomware, before they can cause harm. It involves proactively searching for and detecting threats, and taking steps to prevent or respond to them.

10. How can network segmentation help protect against ransomware?

Answer: Network segmentation can help protect against ransomware by limiting the spread of malware and reducing the attack surface. By isolating sensitive systems and data, organizations can prevent ransomware from spreading and minimize the impact of an attack.